

English 1102 Week 5 Lesson Summary

Research Methodologies and Professional Communication

Overview

This week focused on understanding different research methodologies (qualitative, quantitative, and mixed-methods), determining the appropriate research approach for a study, and drafting professional emails to communicate research ideas effectively.

Key Learning Objectives

- Identify and describe qualitative, quantitative, and mixed-methods research approaches
 - Determine the appropriate research methodology for a study
 - Justify methodology choices based on research goals
 - Draft professional emails with clarity, appropriate tone, and conciseness
 - Apply persuasive communication techniques in academic contexts
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Main Topics Covered

1. Understanding Research Methodologies

Quantitative Research

What It Is:

- Focuses on numerical data and statistical analysis
- Uses measurable variables
- Tests hypotheses and identifies patterns
- Answers "how much" or "how many" questions

When to Use:

- Need to measure relationships between variables
- Want to test specific hypotheses
- Require statistical evidence
- Need generalizable results from large samples

Methods:

- Surveys with closed-ended questions

- Experiments with controlled variables
- Statistical analysis
- Standardized measurements

Strengths:

- Objective and reliable
- Can be replicated
- Allows for statistical comparison
- Clear, measurable results

Example Research Questions:

- "Does increased social media usage correlate with higher anxiety levels?"
 - "How much time do students spend on homework daily?"
 - "What percentage of employees use AI tools at work?"
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Qualitative Research**What It Is:**

- Focuses on understanding experiences, meanings, and perspectives
- Uses non-numerical data (words, images, observations)
- Explores "why" and "how" questions
- Provides rich, detailed descriptions

When to Use:

- Need to understand experiences or motivations
- Exploring new or complex phenomena
- Want detailed, contextual information
- Studying behaviors or cultures

Methods:

- In-depth interviews
- Focus groups
- Observations
- Open-ended surveys
- Content analysis

Strengths:

- Provides deep understanding
- Captures complexity and context
- Flexible and exploratory
- Reveals unexpected findings

Example Research Questions:

- "How do students experience stress during exams?"
 - "What motivates people to change their eating habits?"
 - "Why do employees resist using new technology?"
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Mixed-Methods Research**What It Is:**

- Combines both quantitative and qualitative approaches
- Uses numerical data AND narrative/descriptive data
- Provides comprehensive understanding
- Addresses multiple research questions

When to Use:

- Need both statistical evidence AND detailed explanations
- Want to validate findings through multiple data sources
- Studying complex phenomena requiring different perspectives
- One method alone would be insufficient

Methods:

- Surveys with both closed and open-ended questions
- Experiments followed by participant interviews
- Statistical analysis combined with case studies
- Quantitative measurements with qualitative observations

Strengths:

- Most comprehensive approach
- Provides both breadth and depth
- Can explain unexpected quantitative findings
- Triangulates data for stronger conclusions

Example Research Questions:

- "How does AI assistance affect software development time (quantitative), and what are developers' experiences using these tools (qualitative)?"
 - "What is the correlation between study hours and grades (quantitative), and why do students choose their study methods (qualitative)?"
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2. Choosing the Right Methodology

Decision Framework:

1. What is your research question?

- "How much/many?" → Quantitative
- "Why/how?" → Qualitative
- Both types of questions? → Mixed-methods

2. What kind of data do you need?

- Numbers, measurements, statistics → Quantitative
- Experiences, meanings, descriptions → Qualitative
- Both → Mixed-methods

3. What is your research goal?

- Test hypothesis, find patterns → Quantitative
- Explore experiences, understand context → Qualitative
- Comprehensive understanding → Mixed-methods

4. What resources do you have?

- Large sample, statistical tools → Quantitative works well
 - Time for interviews, smaller sample → Qualitative works well
 - Sufficient time and resources → Mixed-methods possible
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3. Professional Email Writing

Key Components of a Professional Academic Email

Structure:

1. **Subject Line** - Clear and specific
2. **Greeting** - Formal ("Dear Dr. Smith," or "Dear Professor Johnson,")
3. **Introduction** - State purpose immediately
4. **Body** - Explain research idea and methodology
5. **Request** - Clearly state what you need
6. **Closing** - Thank recipient, professional sign-off

7. **Signature** - Your name and contact info

Tone Considerations:

- **Formal but not stiff** - Professional yet approachable
- **Respectful** - Acknowledge recipient's expertise and time
- **Clear** - Get to the point quickly
- **Concise** - Respect the reader's time (200-350 words ideal)
- **Confident** - Show you've thought through your ideas

Common Mistakes to Avoid:

- Casual greetings ("Hey," "Hi there")
 - Being too vague about your request
 - Overly long emails that bury the main point
 - Forgetting to explain WHY you're contacting this person
 - No clear call to action
 - Missing professional closing
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4. Justifying Methodology Choices

What to Include in Your Justification:

1. Explain your research goals

- What do you want to discover?
- What questions need answering?

2. Connect methodology to goals

- How does this method help achieve your goals?
- What specific advantages does it offer?

3. Acknowledge what the method provides

- Quantitative: measurable data, statistical relationships, patterns
- Qualitative: deep understanding, context, participant perspectives
- Mixed-methods: comprehensive view, multiple data types

4. Reference scholarly support

- Cite research methodology sources
- Show you understand methodological principles

Example Justification: "A quantitative approach is most appropriate for this study because it allows for statistical analysis of the relationship between social media usage time and sleep quality (Serdikoff,

2021). Using surveys with standardized sleep quality measures will provide measurable data that can identify clear patterns and correlations across a large sample of college students."

Persuasive Communication Techniques

Key Principles for Academic Persuasion

1. Establish Credibility

- Reference relevant readings and research
- Show you understand the topic
- Use appropriate academic language

2. Provide Clear Reasoning

- Explain your logic step-by-step
- Connect claims to evidence
- Show cause-and-effect relationships

3. Anticipate Questions

- Address potential concerns proactively
- Explain why other approaches won't work as well
- Show you've considered alternatives

4. Use Evidence

- Cite scholarly sources
 - Reference research methodology principles
 - Support claims with examples
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Week 5 Assignment Application

Discussion Post Requirements:

- Draft professional email to potential advisor
- Explain research idea clearly
- Identify appropriate methodology (quantitative, qualitative, or mixed-methods)
- Justify why this approach fits the research goals
- 200-350 words (not including references)
- Include in-text citations and reference list

Success Criteria: ✓ Professional email format and tone ✓ Clear research topic and question ✓ Specific methodology named and explained ✓ Strong justification with scholarly support ✓ Proper citations and references ✓ Within word count limits

Required Readings Summary

Weaver et al. (2016) - Chapter 3.8:

- Persuasive writing techniques
- Building effective arguments
- Rhetorical strategies

Serdikoff (2021) - Chapter 7, Section 36:

- Qualitative research explained
- Overview of quantitative research
- When to use each approach

National University (2024):

- Quantitative research questions
- Sampling methods
- Research design principles

Ashman et al. (2021):

- Examples of persuasive messages
- Professional email formatting
- Effective business communication

Last, S. (n.d.):

- Technical writing principles
- Persuasive communication strategies

Key Takeaways

1. **Different methodologies serve different purposes** - choose based on research goals
 2. **Quantitative** = numbers, measurements, patterns, "how much"
 3. **Qualitative** = experiences, meanings, context, "why/how"
 4. **Mixed-methods** = comprehensive, combines both approaches
 5. **Professional communication** requires appropriate tone, clarity, and conciseness
 6. **Always justify methodology choices** with clear reasoning and scholarly support
 7. **Email format matters** in academic and professional contexts
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Real-World Application

These skills apply to:

- Writing thesis proposals
- Communicating with professors and advisors
- Applying for research positions
- Developing research projects
- Professional workplace communication
- Graduate school applications
- Career networking

Remember: The best research methodology is the one that best answers your specific research question and meets your research goals.